

Royce Nguyen

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science in Data Science | GPA: 3.84/4

Expected June 2027

Relevant Coursework: Probabilistic Modeling and Machine Learning, Data Visualizations, Practice of Data Science, Relational Data Management, Machine Learning: Learning Algorithms, Product Marketing and Management

EXPERIENCE

Data Analyst Intern

June 2025 – August 2025

ECGO

Atlanta, GA

- Used SQL/PostgreSQL to analyze 30,000+ recycling submissions, identifying key user behavior segments and patterns that informed retention and engagement strategy.
- Developed and trained a computer vision model to detect brand names from user-submitted recycling photos achieving 70% accuracy, reducing manual review effort.
- Translated analytical findings into prioritized experiments with UX, Sales, Marketing, and Analytics teams, driving product and campaign changes based on user data.

Data Scientist

February 2024 – February 2025

Biokind Analytics

San Diego, CA

- Analyzed 20,000+ donor records for nonprofit leaders, identifying actionable patterns to guide outreach and fundraising decisions.
- Prompt engineered an LLM workflow to generate personalized letters, reducing manual drafting time by 50%.
- Built an automated data cleaning pipeline and live updating web dashboard that delivered real-time user metrics to executives.

PROJECTS

Scikit-Learn: Under the Hood | Python, FastAPI, React, D3.js

October 2025 – December 2025

- Deployed a full-stack learning app (FastAPI + React) on Render that visualizes ML algorithms with step-by-step execution traces, helping students understand model behavior beyond black-box outputs.
- Built interactive D3.js/Plotly visualizations that respond in real-time to parameter changes, breaking down complex algorithm formulas.
- Implemented RESTful API endpoints to serve algorithm state and computation steps as JSON for frontend.

League of Legends Tower Analysis | Python, Scikit-learn, Matplotlib, Pandas

March 2025 – April 2025

- Engineered predictive features from 150k+ League of Legends match records, using EDA and hypothesis testing to identify early-game indicators that correlate with late-game tower advantages.
- Developed an end-to-end machine learning pipeline to predict a team's end-of-match tower count, achieving an R^2 of 0.86.
- Conducted fairness analysis using permutation tests to verify consistent model performance across a diverse set of teams.

Does National Debt Determine Quality of Life? | Python, Scikit-learn, Pandas

January 2025 – April 2025

- Led a 5 person team analyzing World Happiness Report data across 150+ countries, investigating how GDP, life expectancy, and social support correlate with national well-being scores.
- Built a Random Forest Regression model predicting national happiness scores ($R^2 = 0.85$) from socioeconomic indicators, highlighting the strongest predictors of well being.

TECHNICAL SKILLS

Languages: Python, Java, C#, R, SQL, JavaScript, HTML, CSS, TypeScript

Developer Tools: AWS, Unity, Git, Jupyter Notebook, Microsoft Excel, Tableau

Frameworks/Libraries: Django, Pandas, NumPy, Plotly, Scikit-learn, Seaborn, BeautifulSoup4, PyTorch, React